

SOFTWARE REQUIREMENTS SPECIFICATION

Warehouse Inventory & Pallet Location Tracker

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Problem Statement #27 — Warehouse Inventory & Pallet Location Tracker

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements and use-case behaviour for the Warehouse Inventory & Pallet Location Tracker. The system tracks 3D shelf pallet positions, enforces maximum shelf/rack weight limits, and logs stock movements through barcode/RFID scanning.

1.2 Scope

The scope covers pallet placement validation, pallet inventory recording, pallet location search, stock-movement logging, rack occupancy and pallet-weight monitoring, high-speed SKU lookup, and preservation of confirmed stock-movement records.

1.3 Stakeholders / Actors

Actor	Description
Warehouse Operator	Performs day-to-day inventory operations such as scanning pallets, placing pallets in racks, moving pallets, a
Logistics Supervisor	Monitors warehouse inventory, rack occupancy, pallet weights, and movement information.
Barcode/RFID Scanner	Provides pallet identification information to the system when inventory is scanned during storage, removal, or

2. Overall Description

2.1 Product Perspective

The system supports warehouse inventory operations involving pallet identification, 3D shelf placement, rack capacity validation, inventory lookup, occupancy monitoring, and movement traceability. The Barcode/RFID Scanner acts as an external device supplying pallet identification information.

2.2 Main Product Functions

- Validate pallet placement against rack maximum load capacity.
- Record pallet SKU, quantity and 3D shelf location through barcode/RFID scanning.
- Search for a pallet by SKU and display its current 3D shelf location.
- Record confirmed pallet placements, removals and movements.
- Display rack occupancy and pallet weight information to the Logistics Supervisor.
- Meet the specified SKU lookup performance requirement.
- Preserve confirmed stock-movement records without loss during normal operation.

3. Functional Requirements

ID	Description	Priority
FR-001	The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage.	High
FR-002	The system shall allow a Warehouse Operator to record a pallet's SKU, quantity, and 3D shelf location using barcode/RFID scanning.	High
FR-003	The system shall allow authorized users to search for a pallet using its SKU and display its current 3D shelf location.	High
FR-004	The system shall record every confirmed pallet placement, removal, or movement with the pallet identifier, previous location, new location, and timestamp.	Medium
FR-005	The system shall allow a Logistics Supervisor to view current rack occupancy and pallet weight information for warehouse inventory.	Medium

4. Non-Functional Requirements

ID	Category	Requirement	Priority
NFR-001	Performance	The warehouse item lookup service shall locate any SKU pallet coordinate across 100,000 items in under 100 ms.	High
NFR-002	Reliability	The system shall preserve all confirmed stock-movement records without loss during normal system operation.	High

5. Use Cases

ID	Use Case	Primary / Supporting Actor	Description
UC-01	Validate Rack Capacity	Warehouse Operator	Validate proposed pallet placement against the destination rack's remaining capacity.
UC-02	Record Pallet Inventory	Warehouse Operator + Barcode/RFID Scanner	Record a pallet's SKU, quantity and 3D shelf location, validate rack capacity, update occupancy.
UC-03	Search Pallet Location	Authorized User	Search for a pallet by SKU and display its current 3D shelf location.
UC-04	Log Stock Movement	Warehouse Operator / System	Record confirmed pallet placement, removal or movement with required movement details.
UC-05	View Rack Occupancy	Logistics Supervisor	View current rack occupancy and pallet weight information.

5.1 UC-02 — Record Pallet Inventory

Primary Actor: Warehouse Operator

Supporting Actor: Barcode/RFID Scanner

Brief Description: The Warehouse Operator records a pallet's SKU, quantity and 3D shelf location using barcode/RFID scanning. Before confirming placement, the system validates the destination rack's maximum load capacity and logs the confirmed stock movement.

Preconditions

- The Warehouse Operator is authorized to perform inventory operations.
- The pallet has a valid barcode/RFID identifier.
- The destination rack and shelf location exist in the warehouse system.
- The barcode/RFID scanner and inventory system are operational.

Postconditions — Success

- The pallet's SKU, quantity and 3D shelf location are recorded.
- Rack occupancy information is updated.
- The confirmed stock movement is logged with pallet identifier, previous location, new location, timestamp, and user/device identifier.

Postconditions — Failure

- The pallet is not assigned to the proposed location.
- Inventory information remains unchanged.
- The rejected movement is not recorded as a completed stock movement.

Main Success Scenario

1. Warehouse Operator selects “Record Pallet Inventory.”
2. System requests the pallet's barcode/RFID information.
3. Warehouse Operator scans the pallet.
4. System retrieves the pallet's SKU and existing inventory information.
5. Warehouse Operator enters or selects the pallet quantity and destination 3D shelf coordinates.
6. System validates the proposed pallet placement against the destination rack's maximum load capacity.
7. System confirms that sufficient rack capacity is available.
8. System records the pallet's SKU, quantity and new 3D shelf location.
9. System updates rack occupancy information.
10. System logs the confirmed stock movement with the required movement details.
11. System displays “Pallet Inventory Recorded Successfully.”
12. The use case ends successfully.

Alternate Flow — Rack Capacity Exceeded

1. The system determines that the destination rack does not have sufficient remaining capacity.
2. The system rejects the proposed placement.
3. The system displays “Insufficient Rack Capacity.”
4. The Warehouse Operator is prompted to select another valid rack or shelf location.
5. If another location is selected, the process repeats capacity validation.
6. If the operator cancels the operation, the process ends without updating the pallet location or recording the movement.

6. Acceptance Criteria

ID	Acceptance Criteria
FR-001	Pass: pallet is assigned only when the selected rack has sufficient remaining capacity. Fail: placement is rejected when the rack capacity is exceeded.
FR-002	Pass: after a valid scan, SKU, quantity and shelf coordinates are recorded. Fail: no stock record is created for an invalid scan.
FR-003	Pass: valid SKU returns current shelf, row and level coordinates. Fail: invalid/unavailable SKU produces an appropriate “Pallet Not Found” message.
FR-004	Pass: completed stock movement creates a log containing all required fields. Fail: movement is not considered complete if its log entry is missing.
FR-005	Pass: supervisor can select a rack and view occupied capacity and stored pallet information. Fail: incomplete/unavailable rack data is displayed.
NFR-001	Pass: benchmarking confirms SKU lookup completes in less than 100 ms across 100,000 bins under the specified test load. Fail: lookup time exceeds 100 ms.
NFR-002	Pass: after at least 1,000 consecutive valid movement transactions, all 1,000 corresponding records can be retrieved. Fail: any confirmed movement transaction is lost.

7. Requirements Traceability

The requirements are traced to the project epics and verification test cases maintained in the Requirements Traceability Matrix (RTM). The seven documented project bugs are also linked to the corresponding requirements in the RTM.

Requirement	Epic	Test Case	Defect
FR-001	EPIC-01	TC-001	BUG-01
FR-002	EPIC-02	TC-002	BUG-02
FR-003	EPIC-03	TC-003	BUG-03

Requirement	Epic	Test Case	Defect
FR-004	EPIC-04	TC-004	BUG-04
FR-005	EPIC-05	TC-005	BUG-05
NFR-001	EPIC-06	TC-006	BUG-06
NFR-002	EPIC-07	TC-007	BUG-07

8. References to Project Artifacts

- Requirements and use-case documentation prepared for Problem Statement #27.
- UML Use-Case Diagram for the Warehouse Inventory & Pallet Location Tracker.
- Requirements Traceability Matrix: RTM_SimretMKotian_PES1UG24CS456.xlsx.
- Kanban, Scrum and Bug Report project-management artifacts.